



▲ FIRE AND ICE

The volcanic island of Iceland has been created by a vast outpouring of basalt lava, from a hotspot under the northern end of the Mid-Atlantic Ridge. Iceland has many active volcanoes, including one that lies beneath a sheet of ice, and water seeping down into the hot rock beneath the island erupts in geysers of superheated water and steam.

SEAMOUNTS AND GUYOTS ►

Most volcanoes that erupt from the ocean floor never reach the surface, as seen here in a sonar image. Some of these seamounts were volcanic islands, but have sunk again. Worn flat on top by waves, they are called guyots after their discoverer, Henry Guyot (1807–1884).

OTHER HOTSPOT ISLANDS



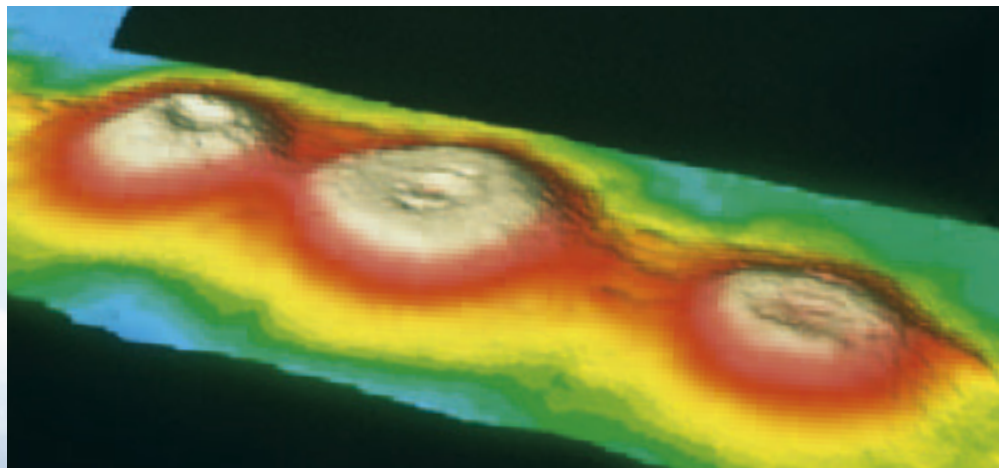
GALÁPAGOS

These volcanic islands formed on a plate moving east over a Pacific hotspot. The islands carried farther to the east are the oldest, while those in the west are still volcanically active. They are famous for their unique wildlife, like these marine iguanas.



EASTER ISLAND

The famous Easter Island statues are carved from volcanic ash erupted by three volcanoes formed over a Pacific hotspot about 3 million years ago. The triangular island they created has since drifted off the hotspot, and the volcanoes are now extinct.



► OCEAN RIDGES

Volcanic activity on the ocean floors has also created many long ridges of seamounts that have become joined together, especially in the Indian Ocean. These form the foundation for coral islands like the Maldives.